CASE STUDY / 01

Cisco DHCP Server Configuration Lab

Cisco Packet Tracer | Local network simulation

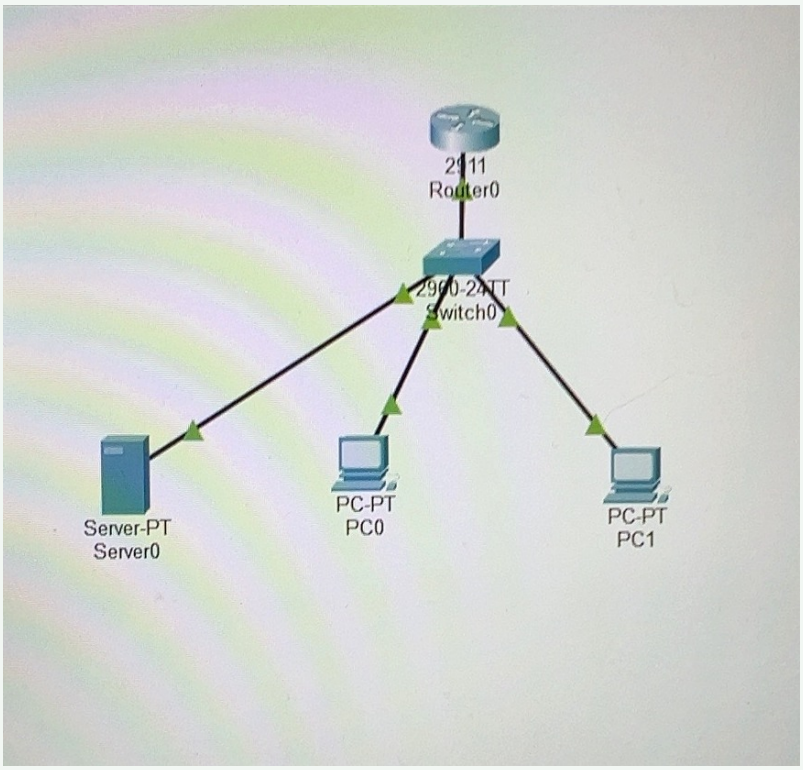
2 DHCP CLIENTS

0% PACKET LOSS

Project overview

Configured a dedicated DHCP server and verified automatic IPv4 addressing on two client computers connected through a Cisco switch. Confirmed local PC-to-PC and default gateway reachability with ping tests.

Network topology

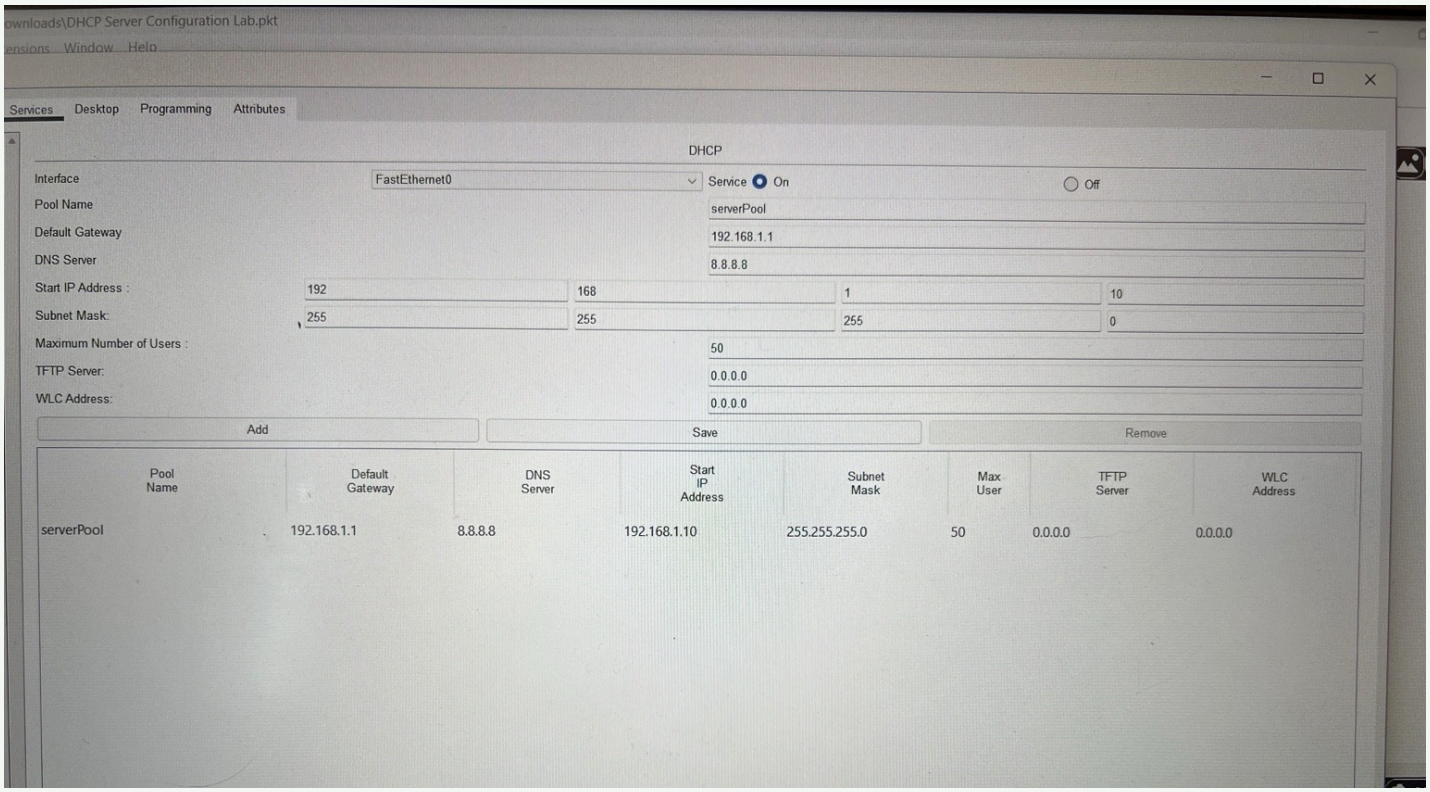


2911 router | 2960-24TT switch | Server-PT | PC0 | PC1

Evidence: Packet Tracer topology screenshot provided by the student.

DHCP service configuration

Server0 / Services > DHCP



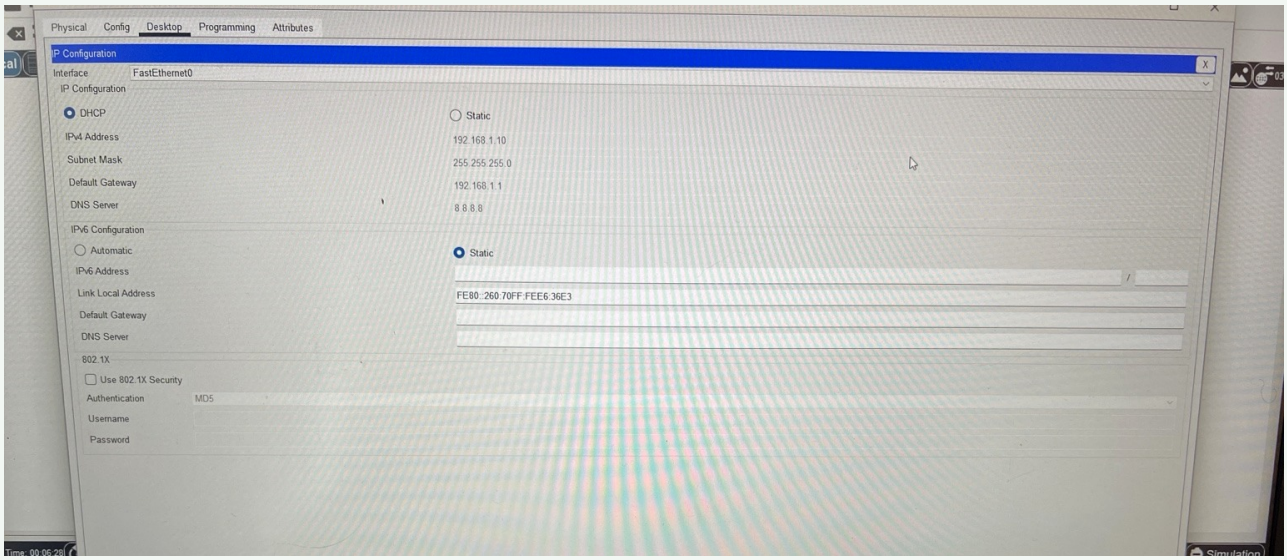
Configured DHCP pool

Parameter	Observed configuration
Service	On
Interface	FastEthernet0
Pool name	serverPool
Starting IPv4 address	192.168.1.10
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server	8.8.8.8
Maximum users	50

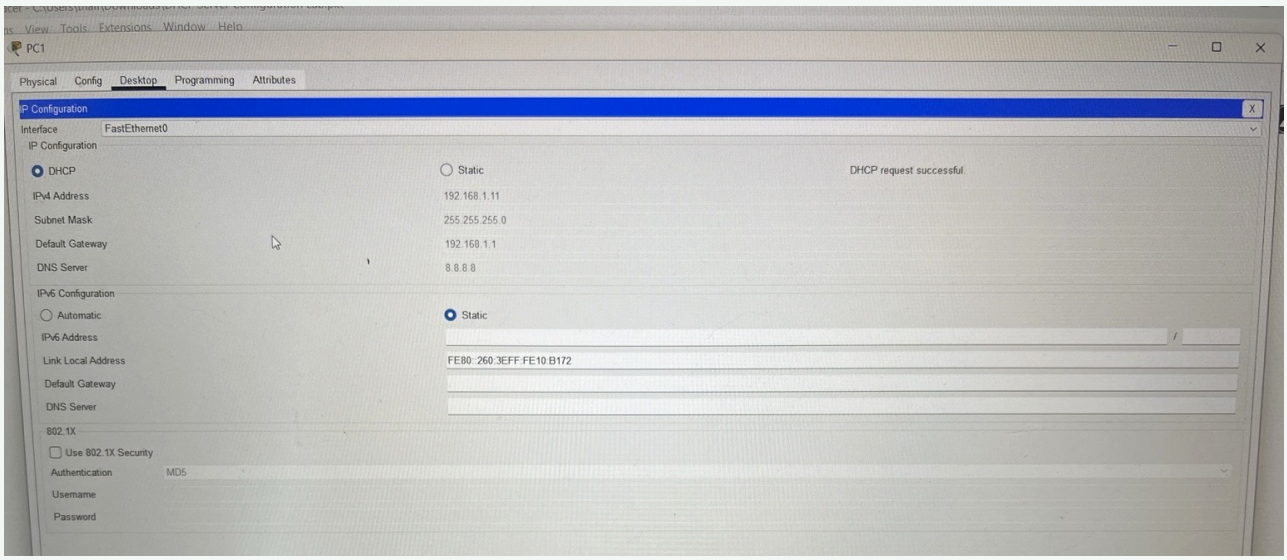
Scope: These settings are confirmed in the Server0 DHCP panel.

DHCP client address assignments

PC0 / Desktop > IP Configuration



PC1 / Desktop > IP Configuration

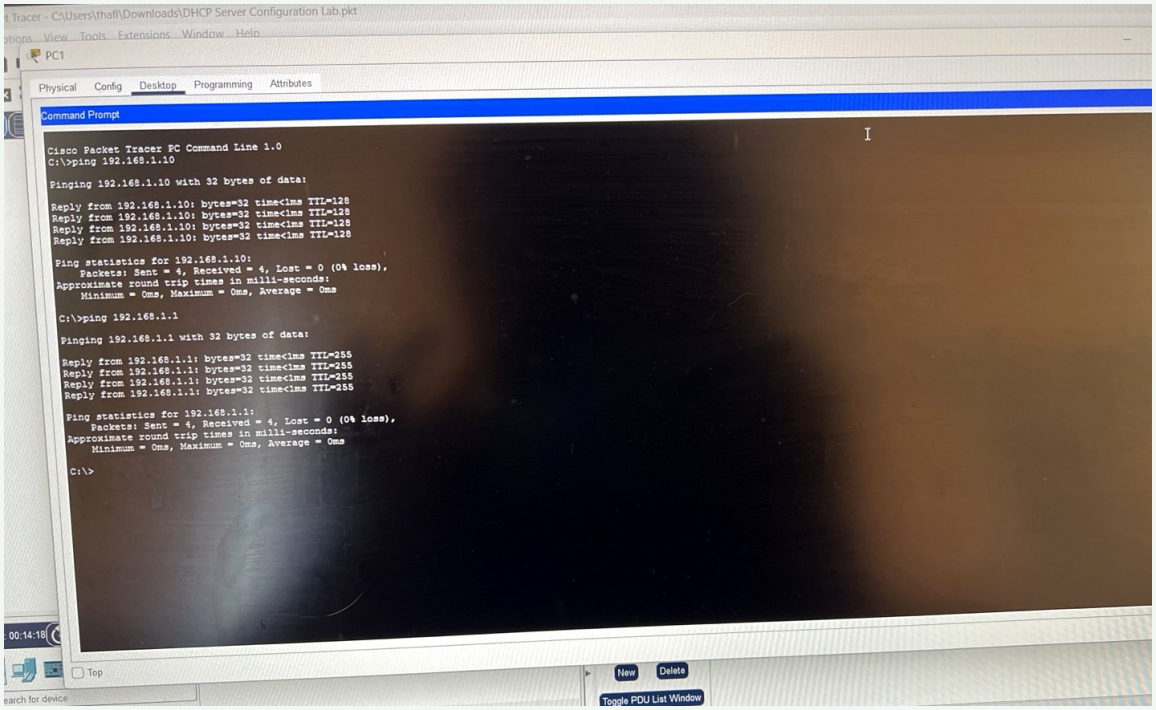


Field	PC0	PC1
IPv4 address	192.168.1.10	192.168.1.11
Subnet mask	255.255.255.0	255.255.255.0
Default gateway	192.168.1.1	192.168.1.1
DNS server	8.8.8.8	8.8.8.8

Result: Both clients obtained unique addresses from the configured DHCP pool.

Connectivity verification and findings

PC1 / Command Prompt



Observed ping tests

Source	Destination	Sent / received	Loss
PC1	PC0 - 192.168.1.10	4 / 4	0%
PC1	Gateway - 192.168.1.1	4 / 4	0%

Outcome

Both PCs received DHCP-assigned IPv4 settings. PC1 successfully reached PC0 and the configured router gateway, each with four of four ping replies and no packet loss.

Testing boundaries

This evidence confirms local IP addressing and reachability within the Packet Tracer simulation. It does not verify public internet connectivity, external DNS resolution, or production deployment.

Portfolio skills demonstrated

DHCP | IPv4 addressing | Router/switch topology | Ping testing | Documentation